

Maharashtra Public Service Commission Assessment

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Assistant Professor Chemistry Group A

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Group Id :	1505618
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Show Attended Group? :	No
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Assistant Professor Chemistry Group A

Section Id :	1505618
Section Number :	1
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Sub-Section Number :	1
Sub-Section Id :	15056113
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 1 Question Id : 150561703 Question Type : MCQ Option Shuffling : No

What are the eigenvalues for Z component of angular momentum operator L_Z ?
(m_l = magnetic quantum number)

Options :

1. $L_Z = m_l \hbar$

2. $L_Z = \frac{\hbar}{2\pi m_l}$

3. $L_Z = m_l \hbar^2$

4. $L_Z = 0$

Which quantum number defines the orientation of orbital in the space around the nucleus ?

Options :

1. Principal quantum number
2. Angular quantum number
3. Magnetic quantum number
4. Spin quantum number

When the particle is present in one-dimensional potential box, the energy of the lowest level ($n = 1$) is called _____.

Options :

1. Kinetic energy
2. Potential energy
3. Zero point energy
4. Ionization energy

Pauli's exclusion principle states that _____.

Options :

1. Electrons occupy orbitals of lowest energy
2. Nucleus of an atom contains no negative charge
3. Electrons move in circular orbits around the nucleus
4. All the four quantum numbers of two electrons in an atom cannot be equal

According to the LCAO-MO theory for H₂, what is the bond order ?

Options :

1. 2
2. 1
3. 0
4. 3

All symmetrical linear molecules belong to the point group _____.

Options :

1. D_{∞h}
2. I_h
3. T_d
4. C_{3v}

The correct option to complete the following irreversible representation of C_{2v} point group is _____.

	$\hat{E}$	$\hat{E}_2$	$\hat{\sigma}_V$	$\hat{\sigma}_{V_1}$
A ₁	1	1	1	1
A ₂	1	1	-1	-1
B ₁				
B ₂	1	-1	-1	1

Options :

1. 1, -1, 1, -1
2. -1, 1, -1, 1
3. -1, -1, 1, 1
4. 1, 1, -1, -1

Which of the following nucleus does not show nuclear magnetic resonance ?

Options :

1. ^{13}C
2. ^{15}N
3. ^{16}O
4. ^{19}F

The relative intensities of the spectral lines in rotational spectra of diatomic molecules is given by _____.

Options :

1. $\overline{\Delta\nu} = 4B (J + 2)$
2. $\overline{\Delta\nu} = 2B (J + 2)$
3. $\overline{\Delta\nu} = 4B (J + 1)$
4. $\overline{\Delta\nu} = 2B (J + 1)$

Which spectroscopic technique is used to identify functional groups in an organic compound ?

Options :

1. UV – Vis
2. IR
3. NMR
4. Mass

Question Number : 11 Question Id : 150561713 Question Type : MCQ Option Shuffling : No

Amount of heat required to raise the temperature of one mole of the system by 1°C is known as _____.

Options :

1. Heat capacity
2. Molar heat capacity
3. Specific heat
4. Chemical potential

Question Number : 12 Question Id : 150561714 Question Type : MCQ Option Shuffling : No

A one component system in single phase, is _____.

Options :

1. invariant
2. monovariant
3. bivariant
4. trivariant

Question Number : 13 Question Id : 150561715 Question Type : MCQ Option Shuffling : No

With increase in temperature, the value of the partition function will _____.

Options :

1. decrease
2. increase
3. remain the same
4. go through a maximum

The entropy of perfectly pure crystalline substance is zero at absolute zero temperature. This is the statement of _____.

Options :

1. First law of thermodynamics
2. Second law of thermodynamics
3. Third law of thermodynamics
4. Zeroth law of thermodynamics

For the isothermal compression of an ideal gas, the reversible work done is given as $W_{\text{rev}} = -nRT \ln V_2/V_1$. The correct option among the following is _____.

Options :

1. $V_2 > V_1$ and work is done on the gas
2. $V_2 > V_1$ and work is done by the gas
3. The external pressure is infinitesimally lower than the pressure of the gas
4. $V_2 < V_1$ and work is done on the gas

The correct statement about change in entropy (ΔS) among the following is _____.

Options :

1. For a reversible process, $\Delta S_{\text{universe}} > 0$
2. For an irreversible process, $\Delta S_{\text{universe}} = 0$
3. Entropy of the universe increases in a spontaneous process
4. All spontaneous processes are reversible

Which of the following statements is correct for elementary reaction ?

Options :

1. It will have a single intermediate
2. It occurs in a single step
3. Molecularity of an elementary reaction depends on concentration
4. All elementary reactions are first order reactions

In a potential energy diagram saddle point represents _____.

Options :

1. The reactants
2. The products
3. An intermediate
4. The transition state

A large negative value of entropy (ΔS°)[#] in the Eyring equation indicates _____.

Options :

1. Less ordered activated complex than the reactants
2. Highly ordered activated complex than the reactants
3. High values of steric factor
4. Higher degrees of freedom than the reactants

Which of the following statement is not correct for order of a reaction ?

Options :

1. Order of reaction can be determined experimentally
2. It is sum of the powers of concentration terms in the rate law expression
3. It doesn't necessarily depend on the stoichiometric coefficients
4. Order of a reaction cannot be fractional

Question Number : 21 Question Id : 150561723 Question Type : MCQ Option Shuffling : No

The molecularity of the reaction cannot be _____.

Options :

1. 1
2. 2
3. 1.5
4. 3

Question Number : 22 Question Id : 150561724 Question Type : MCQ Option Shuffling : No

The correct statement about colloids among the following is _____.

Options :

1. Milk is a colloidal dispersion of water in fat
2. Colloids can be stabilised by the repulsive interactions of the electrical double layer
3. Colloids can be stabilised by the attractive interactions of the electrical double layer
4. In colloids, the dispersed phase is the continuous phase

Surface tension of water and benzene at 25°C are 72.9 and 28.9 mN/m respectively. This huge difference can be due to _____.

Options :

1. Lower molar mass of water than that of benzene
2. Lower density of benzene than that of water
3. Presence of both van der Waals bonds and hydrogen bonds in water
4. Aromatic character of benzene

Langmuir while deriving adsorption isotherm did not make the following assumption _____.

Options :

1. The layer of the gas adsorbed on the solid surface is one molecule thick
2. The adsorbed layer is uniform
3. There is no attraction between the adjacent molecules
4. The attraction between adsorbent molecule is extremely large

The correct statement about the Critical Micelle Concentration (CMC) among the following is _____.

Options :

1. Beyond CMC, surface tension increases linearly with the addition of the surfactant
2. At CMC, there is sudden decrease in turbidity due to the formation of light scattering species of small sizes
3. Beyond CMC, there is no further adsorption of the surfactant at the interface
4. At CMC, the equivalent conductivity decreases due to an increase in mass/charge ratio

Which of the following is not a characteristic of chemisorption ?

Options :

1. Reversible in nature
2. Unilayer
3. Highly specific
4. Occurs at high temperature

Which allotropes of carbon consists of spherical molecules composed of 60 carbon atoms forming a truncated icosahedron ?

Options :

1. Graphite
2. Diamond
3. Fullerene
4. Carbon nanotube

Which of the following hydrides shows the strongest basic character in aqueous solution ?

Options :

1. NH_3
2. PH_3
3. AsH_3
4. SbH_3

What is the composition of water gas ?

Options :

1. Mixture of CO, H₂, CH₄ and CO₂
2. Mixture of CO, H₂, CH₄
3. Equimolar mixture of CO and H₂
4. Only H₂ and O₂

Question Number : 30 Question Id : 150561732 Question Type : MCQ Option Shuffling : No

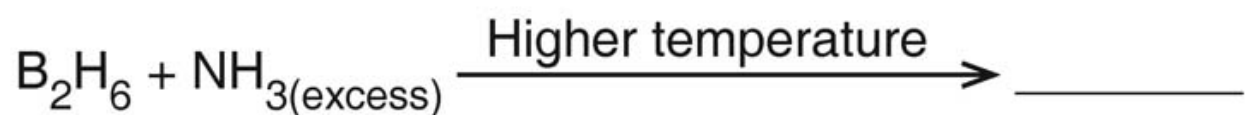
Among the following compounds, which oxoacids of sulphur have sulphur in +III oxidation state ?

Options :

1. Sulphurous acid
2. Dithionous acid
3. Sulphuric acid
4. Peroxo disulphuric acid

Question Number : 31 Question Id : 150561733 Question Type : MCQ Option Shuffling : No

Identify the product.



Options :

1. Boron nitride
2. Diammonia diborane
3. Borazine
4. All of the above

Question Number : 32 Question Id : 150561734 Question Type : MCQ Option Shuffling : No

The nephelauxetic effect is minimum in _____ complexes.

Options :

1. Fluoro
2. Chloro
3. Bromo
4. Iodo

Question Number : 33 Question Id : 150561735 Question Type : MCQ Option Shuffling : No

What is the d^n configuration and the spin multiplicity of the ground state of Ti^{3+} and V^{3+} ?

Options :

1. d^1 ; doublet and d^2 ; triplet
2. d^2 ; doublet and d^1 ; triplet
3. Both (1) and (2)
4. None of the above

Question Number : 34 Question Id : 150561736 Question Type : MCQ Option Shuffling : No

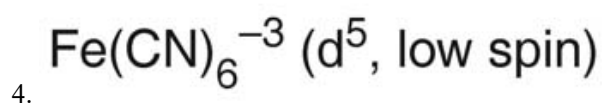
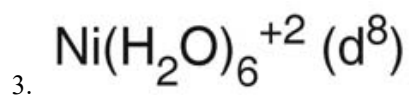
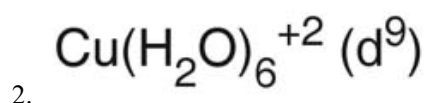
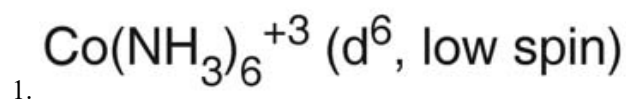
Estimate the reduction in interelectronic repulsion on going from gaseous Rh^{3+} to $[Rh(NH_3)_6]^{3+}$.

Options :

1. ~ 50%
2. 89.72%
3. 18.04%
4. ~ 39%

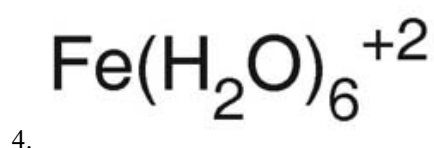
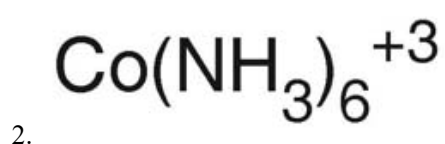
Which of the following octahedral complexes is most likely to exhibit Jahn-Teller distortion ?

Options :



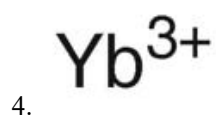
Which of the following complexes will have 4 unpaired electrons ?

Options :



Which of the following is the most paramagnetic ion ?

Options :



Which method is most commonly used for the separation of lanthanides from their mixtures ?

Options :

1. Electrolysis
2. Oxidation with permanganate
3. Precipitation as sulfides
4. Ion-exchange chromatography

For magnetic moment calculation, in the lanthanides the spin contribution (S) and orbital contribution (L) couple together to give a new quantum number (J). Which of the following is correct when the shell is less than half filled ?

Options :

1. $J = L - S$
2. $J = S - L$
3. $J = L + S$
4. $J = LS$

The metal-carbon bond length in metallocenes generally increases with _____.

Options :

1. Decreasing number of unpaired electrons on the metal
2. Increasing number of unpaired electrons on the metal
3. Decreasing metal-ligand orbital overlap
4. Increasing ionic-character of the metal-ligand bond

The conversion of methanol into acetic acid involves _____.

Options :

1. Wacker's process
2. Oxo process
3. Monsanto process
4. Vaska's process

The Fischer-Tropsch synthesis involves _____.

Options :

1. $n\text{CO} + (2n+1) \text{H}_2 \longrightarrow \text{C}_n\text{H}_{2n+2} + n\text{H}_2\text{O}$
2. $\text{H}_2\text{O} + \text{CO} \rightleftharpoons \text{CO}_2 + \text{H}_2$
3. $\text{R}-\text{CH}=\text{CH}_2 + \text{H}_2 + \text{CO} \rightleftharpoons \text{R}-\text{CH}_2-\text{CH}_2-\text{CHO}$
4. All of the above

Which of the following does not obey 18 electron rule ?

Options :

1. $\text{Cr}(\text{CO})_6$
2. $\text{Mn}_2(\text{CO})_{10}$
3. $\text{Fe}(\text{CO})_5$
4. $\text{V}(\text{CO})_6$

Consider the following reaction :

$\text{Mo}(\text{CO})_6 + \text{L} \longrightarrow \text{Mo}(\text{CO})_5\text{L} + \text{CO}$. If the rate of this substitution reaction is independent of the concentration of the incoming ligand L, what is the likely mechanism type ?

Options :

1. Oxidative addition
2. Reductive elimination
3. Dissociative mechanism
4. Associative mechanism

The change ${}_{84}^{215}\text{Po} \longrightarrow {}_{82}^{211}\text{Po}$ requires the emission of _____.

Options :

1. β - Particle
2. γ - Particle
3. α - Particle
4. Positron

The Fissionable material commonly used in atomic bomb is _____.

Options :

1. ${}_{92}^{238}\text{U}$
2. ${}_{92}^{235}\text{U}$
3. ${}_{94}^{239}\text{Pu}$
4. ${}_{90}^{232}\text{Th}$

An important method for fixation of nitrogen is _____.

Options :

1. Frasch's process
2. Deacon's process
3. Solvay's process
4. Haber's process

The green colour of plant leaves is due to the presence of _____.

Options :

1. Iron
2. Calcium
3. Magnesium
4. Zinc

When crystals of NaCl are heated in an atmosphere of sodium vapours, it impart a yellow colour. The reason is _____.

Options :

1. Schottky defect
2. Frenkel defect
3. Metal excess defect
4. Metal deficiency defect

The radius of the Na^+ ion is 102 pm and the radius of the Cl^- ion is 181 pm. Based on the radius ratio rule, what is the predicted coordination number for Na^+ in NaCl crystal ?

Options :

1. 12
2. 8
3. 4
4. 6

Thermogravimetric analysis is not concerned with _____.

Options :

1. Time
2. Temperature
3. Weight
4. Reagent

The capillary constant in Ilkovic equation is given by _____.

Options :

1. $m^{2/3} \cdot t^{1/6}$
2. $m^{1/2} \cdot d^{2/3}$
3. $m^{1/3} \cdot t^{2/3}$
4. $m^{1/4} \cdot d^{1/5}$

McLafferty rearrangement involves _____.

Options :

1. Cleavage of a gamma bond followed by beta hydrogen transfer
2. Cleavage of an alpha bond followed by beta hydrogen transfer
3. Cleavage of a beta bond followed by gamma hydrogen transfer
4. None of the above

Alkali earth metals form M^{2+} ions instead of M^+ ions because _____.

Options :

1. M^{2+} ions carry more charge
2. Their IE_2 values are not different from their IE_1 values
3. M^+ ions do not have stable configuration
4. M^{2+} ions have more hydration energy

If the absorption of light by the coloured solution is 90% then its absorbance is _____.

Options :

1. 10
2. 1
3. 0.1
4. 0.9

The percentage of nitrogen in urea is about _____.

Options :

1. 70
2. 63
3. 47
4. 28

Pure orthophosphoric acid exhibits syrupy nature because it _____.

Options :

1. is triprotic acid.
2. is hygroscopic.
3. contains PO_4 groups connected together by H-bonds.
4. is a strong acid.

The use of hydrogen peroxide in medicine as _____.

Options :

1. Oxidising agent
2. Propellant
3. Antiseptic
4. Lead painting

In the soil, CaCN_2 which is used as a fertiliser, is converted into _____ by atmospheric CO_2 .

Options :

1. Urea
2. NO_2
3. NO
4. CO

The reaction $2\text{NH}_3 + \text{CO}_2 \longrightarrow \text{H}_2\text{NCOONH}_4 \longrightarrow \text{_____} + \text{H}_2\text{O}$.

Options :

1. $\text{NH}_2\text{COONH}_4$
2. H_2NCONH_2
3. $\text{NH}_4\text{COONH}_4$
4. $(\text{NH}_4)_2\text{SO}_4$

The Born-Haber cycle is an application of which fundamental law of thermodynamics ?

Options :

1. The law of conservation of energy
2. Hess's law of constant heat summation
3. The second law of thermodynamics
4. Dalton's law of partial pressures

What is the role of KMnO_4 in acidic medium and what is the reduction product of the permanganate ion ?

Options :

1. Oxidizing agent, Mn^{2+}
2. Oxidizing agent, MnO_2
3. Reducing agent, MnO_2
4. Reducing agent, Mn^{2+}

Alkali metal hydrides react with water to give _____.

Options :

1. Acedic solution
2. Basic solution
3. Neutral solution
4. Hydride ions

An organic molecule is asymmetric if it possesses _____.

- a. A plane of symmetry
- b. A centre of symmetry
- c. An alternating axis of symmetry
- d. None of the above

Options :

1. a and b
2. b and c
3. a, b and c
4. d

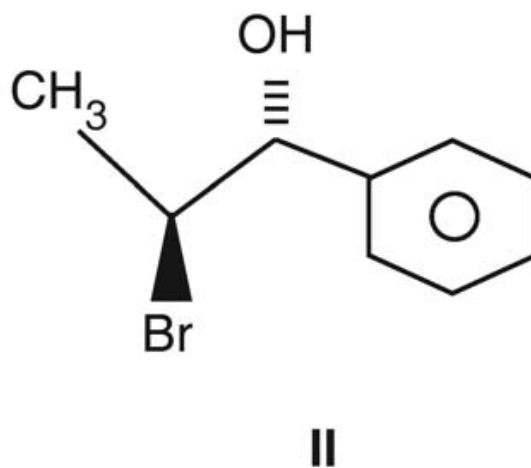
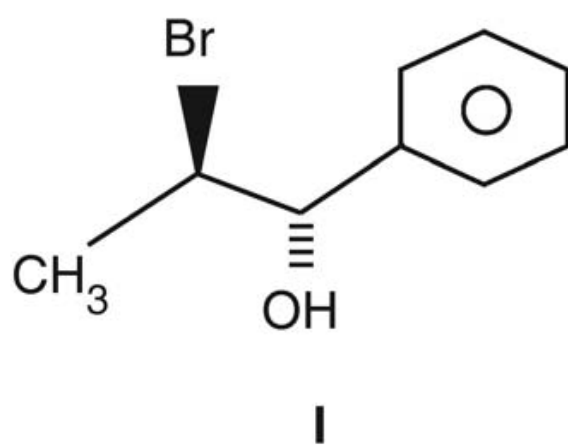
In trans-1, 4-disubstituted cyclohexane, which arrangement of substituents is favoured in the most stable chair conformation ?

Options :

1. Both axial
2. Both equatorial
3. One axial, one equatorial
4. Stability is identical for all conformations

Question Number : 66 Question Id : 150561768 Question Type : MCQ Option Shuffling : No

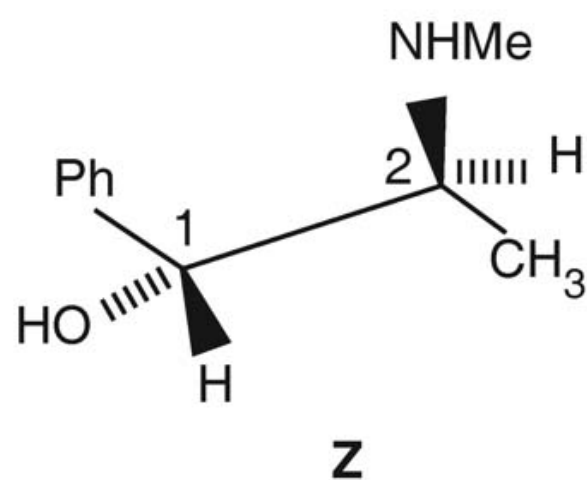
Identify the stereochemical relationship between the following configurational isomers I and II.



Options :

1. Enantiomers
2. Diastereoisomers
3. Epimers
4. Anomers

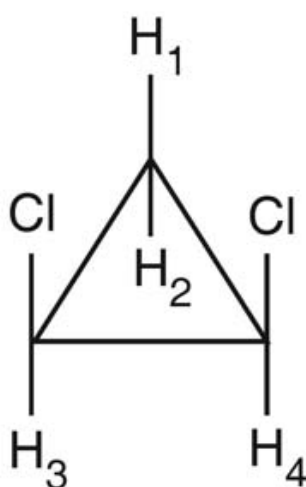
The absolute configurations at C₁ and C₂ in compound Z are _____.



Options :

1. 1R, 2R
2. 1R, 2S
3. 1S, 2S
4. 1S, 2R

Topicity of designated hydrogen atoms in the following molecule is _____.



Options :

1. H₁ and H₂ are diastereotopic while H₃ and H₄ are enantiotopic
2. H₁ and H₂ are enantiotopic while H₃ and H₄ are diastereotopic
3. H₁ and H₂ are homotopic while H₃ and H₄ are enantiotopic
4. H₁ and H₂ are enantiotopic while H₃ and H₄ are homotopic

The rate of hydrolysis of 2-chloroethyl ethyl sulfide is 1000 times faster than n-hexyl chloride in water because of _____.

Options :

1. Inductive effect
2. Neighbouring group participation
3. Mesomeric effect
4. Steric effect

How many isoprene units are present in diterpene ?

Options :

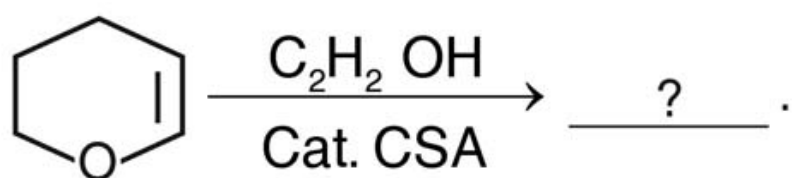
1. 2
2. 3
3. 4
4. 5

The increasing rate of elimination by the E-1 mechanism in case of primary, secondary and tertiary alkyl halide is _____.

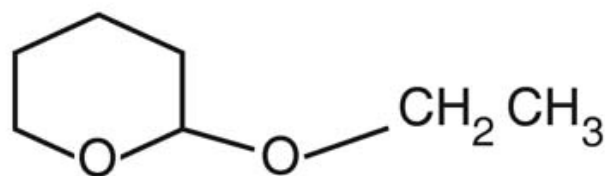
Options :

1. $R_3CX < R_2CHX < RCH_2X$
2. $R_2CHX < RCH_2X < R_3CX$
3. $RCH_2X < R_2CHX < R_3CX$
4. $RCH_2X < R_3CX < R_2CHX$

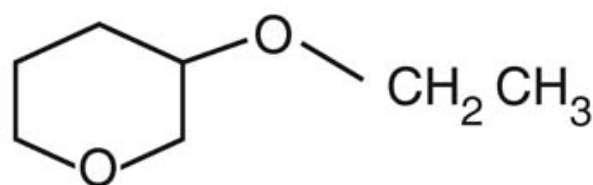
The product obtained in the reaction of dihydropyran with ethyl alcohol in the presence of catalytic amount of camphor sulfonic acid is :-



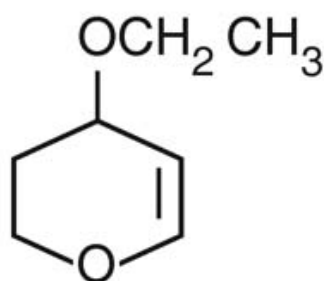
Options :



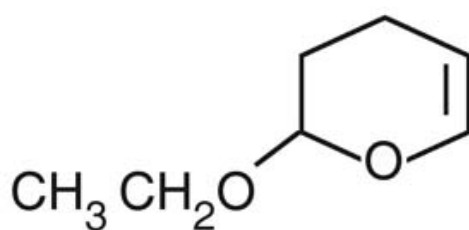
1.



2.



3.



4.

Which statement of the following is true for $\text{S}_{\text{N}}2$ reactions ?

Options :

1. Its transition state involves a carbon atom having five groups

1.

2. Its transition state involves a carbon atom having two groups

2.

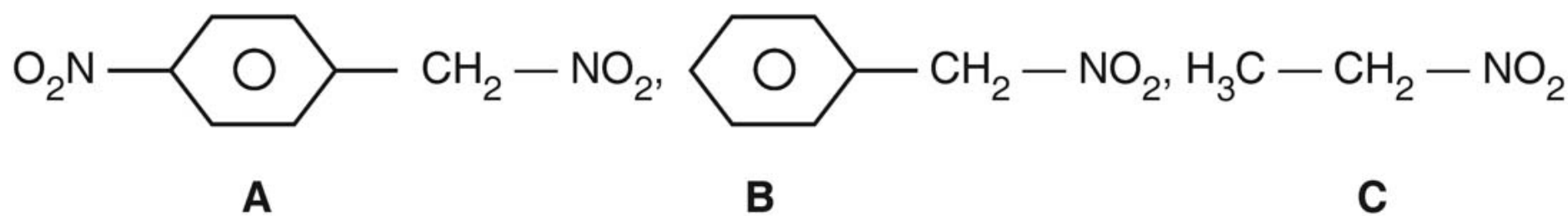
3. Its transition state involves a carbon atom having only three groups

3.

4. None of the above

4.

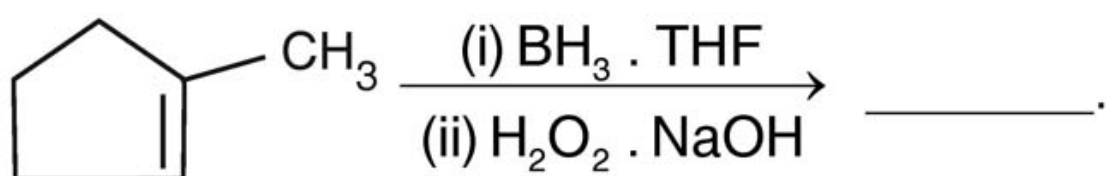
What will be decreasing order of stability of carbanion generated from following molecules ?



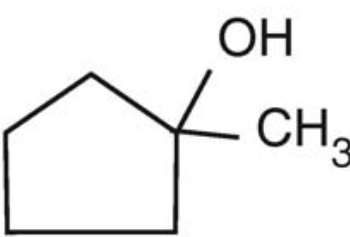
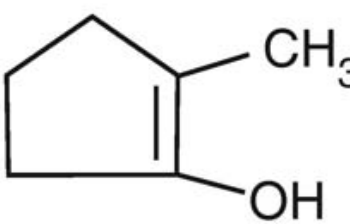
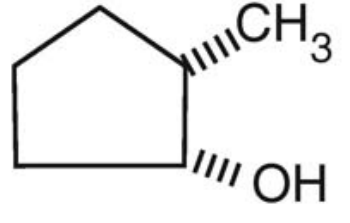
Options :

1. $\text{C} > \text{B} > \text{A}$
2. $\text{A} > \text{B} > \text{C}$
3. $\text{B} > \text{A} > \text{C}$
4. $\text{A} > \text{C} > \text{B}$

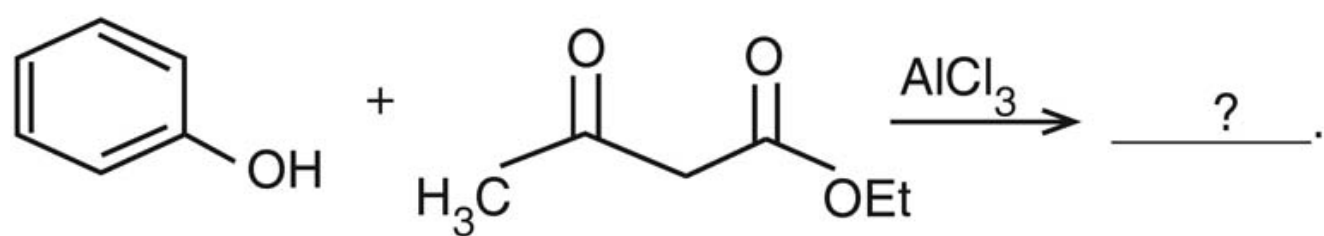
The major product formed in the following reaction is :-



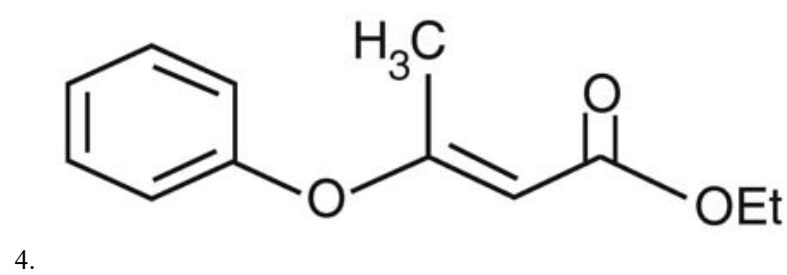
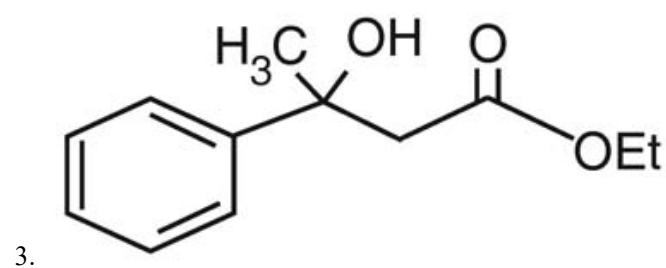
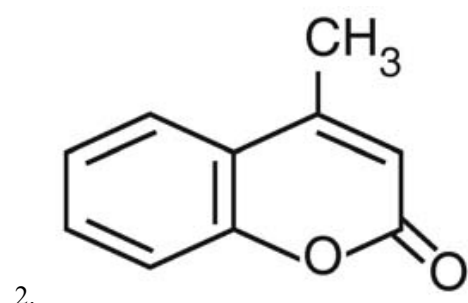
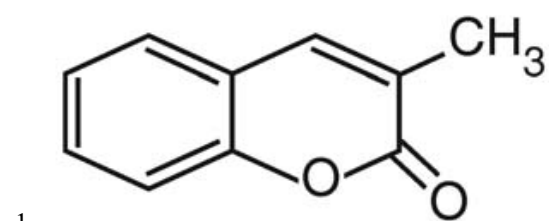
Options :

1. 
2. 
3. 
4. 

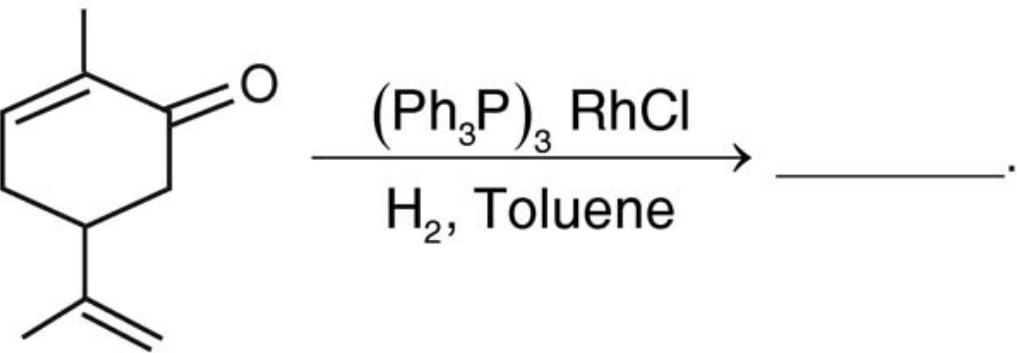
The product in the following reaction is :-



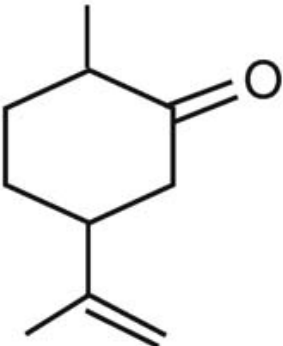
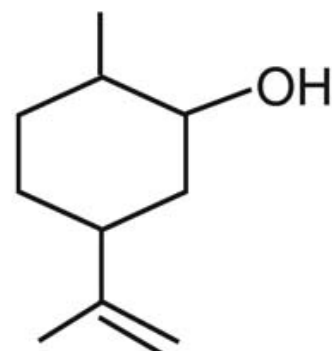
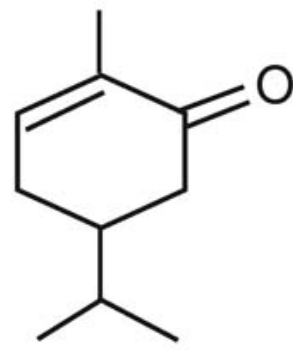
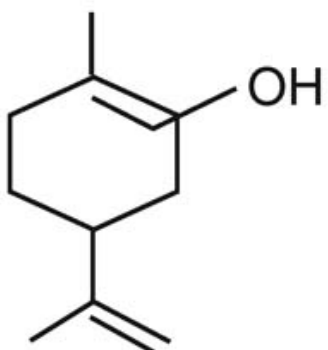
Options :



The major product formed in a following reaction is :-



Options :

1. 
2. 
3. 
4. 

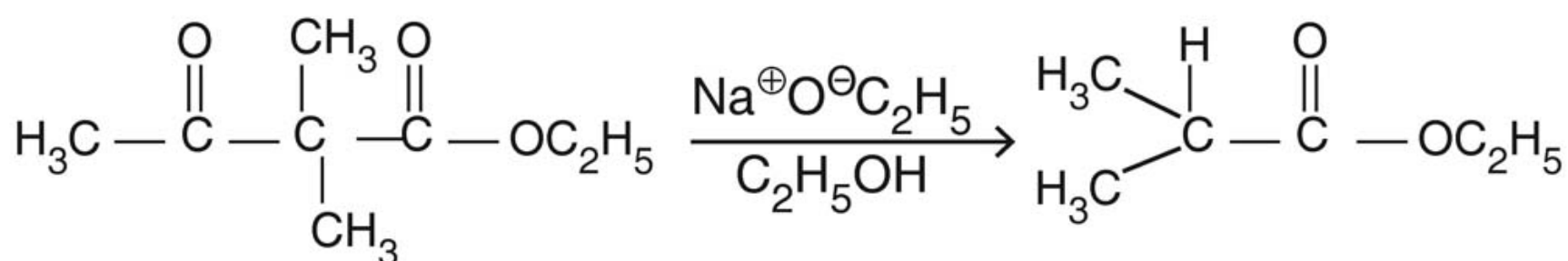
Which of the following principles of green chemistry focuses on maximizing the incorporation of all materials used in the process into final product ?

Options :

1. Inherently safer chemistry
2. Design for energy efficiency
3. Atom economy
4. Less hazardous chemical synthesis

Question Number : 79 Question Id : 150561781 Question Type : MCQ Option Shuffling : No

The name of following reaction is _____.



Options :

1. Claisen reaction
2. Reformatsky reaction
3. Reverse Claisen reaction
4. Reverse Reformatsky reaction

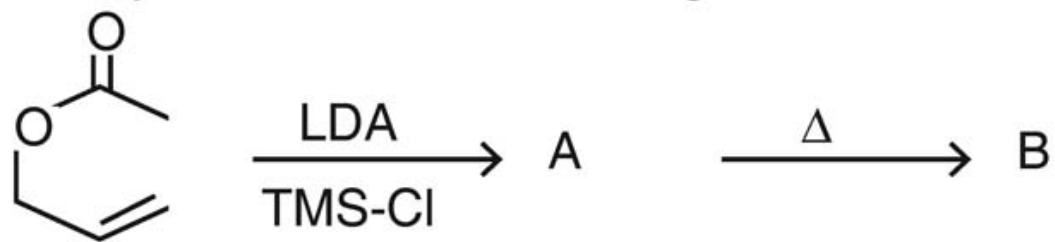
Question Number : 80 Question Id : 150561782 Question Type : MCQ Option Shuffling : No

The oxidation of aliphatic ketones with peroxyacids to form an ester is named as _____.

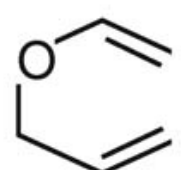
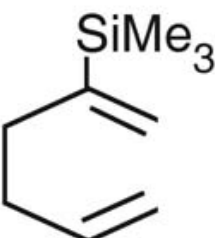
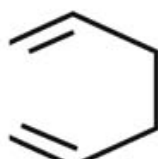
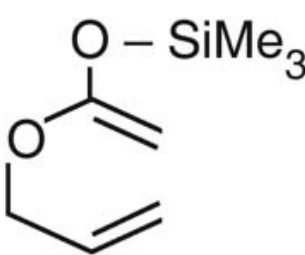
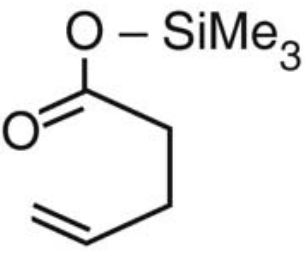
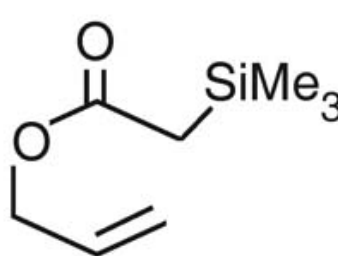
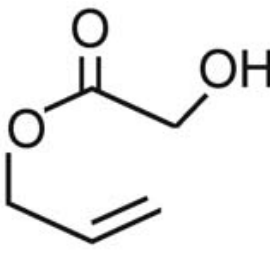
Options :

1. Baeyer Villiger oxidation
2. Haloform reaction
3. Reformatsky reaction
4. Wittig reaction

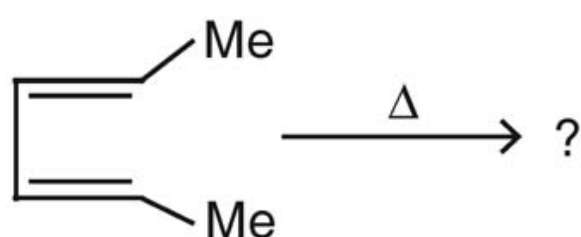
Identify A and B in the following reaction.



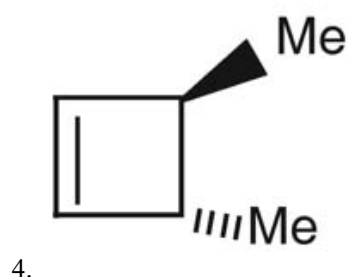
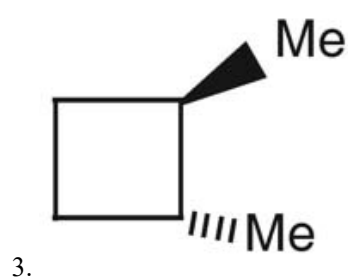
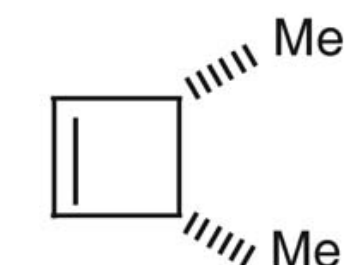
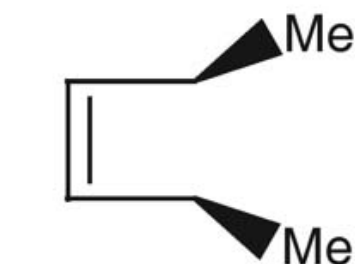
Options :

1. $A =$  $B =$ 
2. $A =$  $B =$ 
3. $A =$  $B =$ 
4. $A =$  $B =$ 

Identify the product in the following reaction.



Options :



Homolytic cleavage of RO — NO bond of nitrite to nitric oxide and alkoxy radical and migration of nitric oxide is known as _____.

Options :

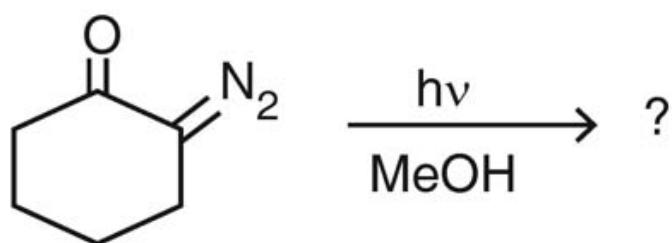
1. Photo fries reaction

2. Barton reaction

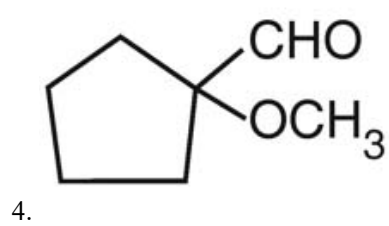
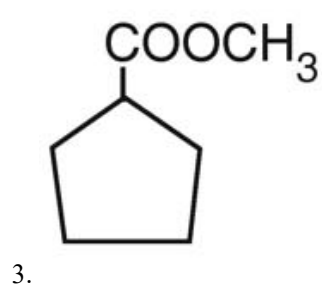
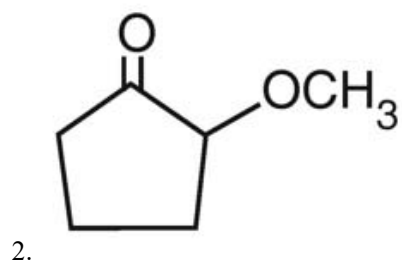
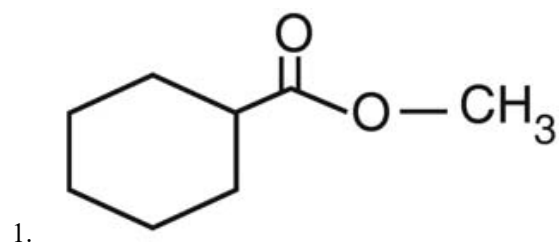
3. Oxime irradiation

4. Hoffmann reaction

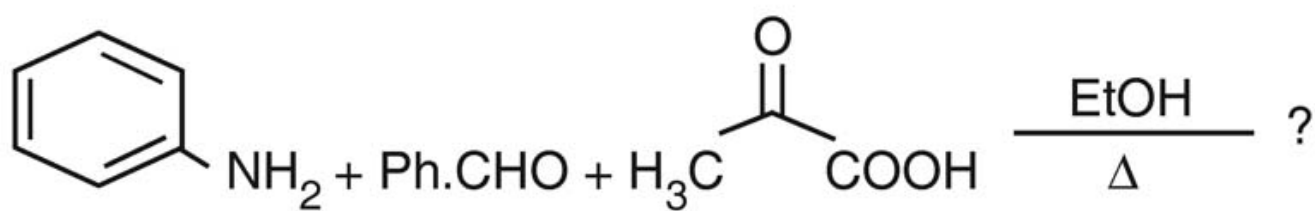
The product obtained in the following reaction is :-



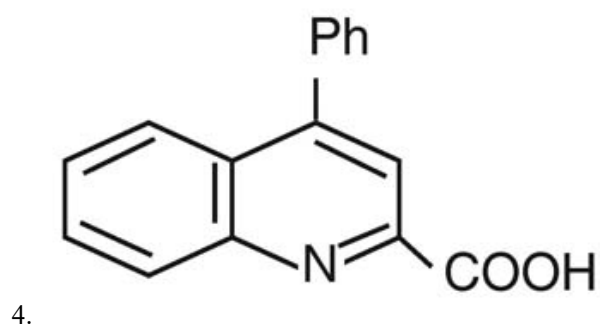
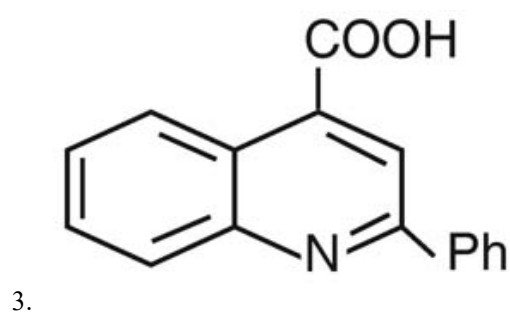
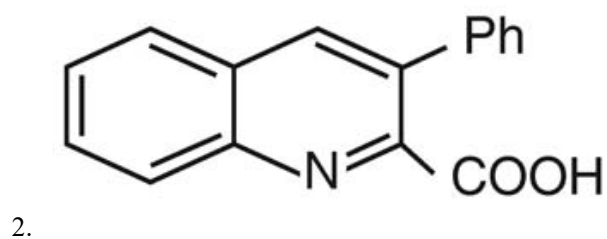
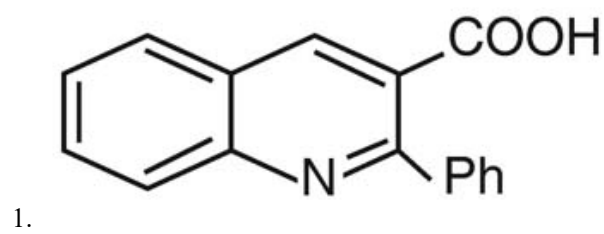
Options :



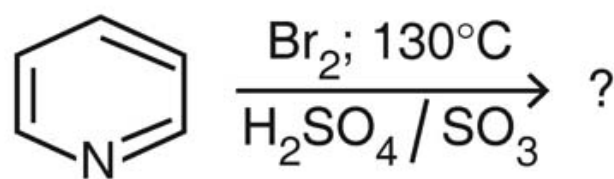
The correct product in the following reaction is :-



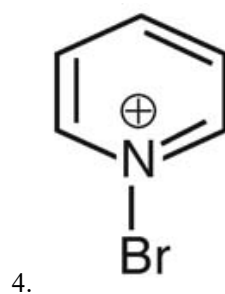
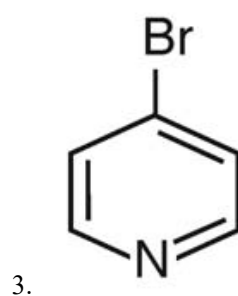
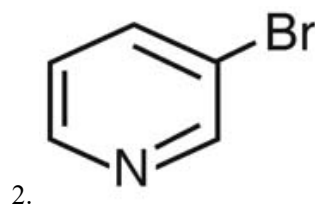
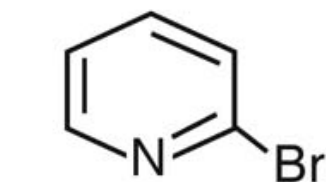
Options :



The major product obtained in the following reaction is :-



Options :



Which of the following shows maximum canonical structures ?

Options :

1. Thiophene

2. Furan

3. Pyrrole

4. Quinoline

Which reagent is most commonly used for coupling in peptide synthesis ?

Options :

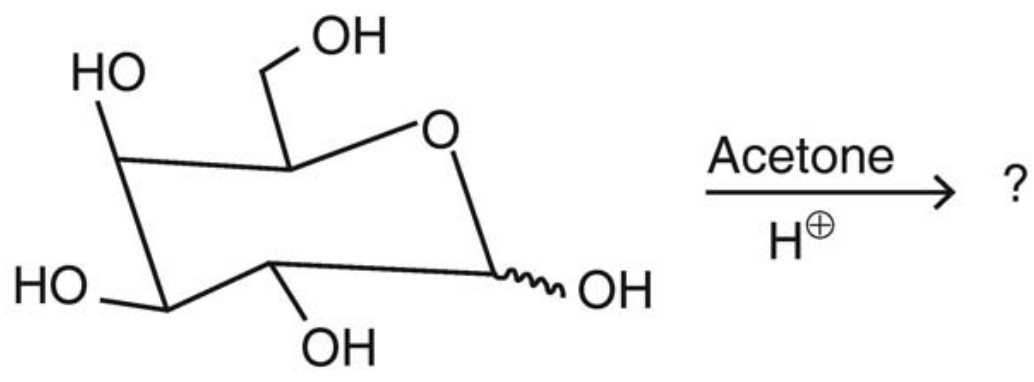
1. DCC

2. DDQ

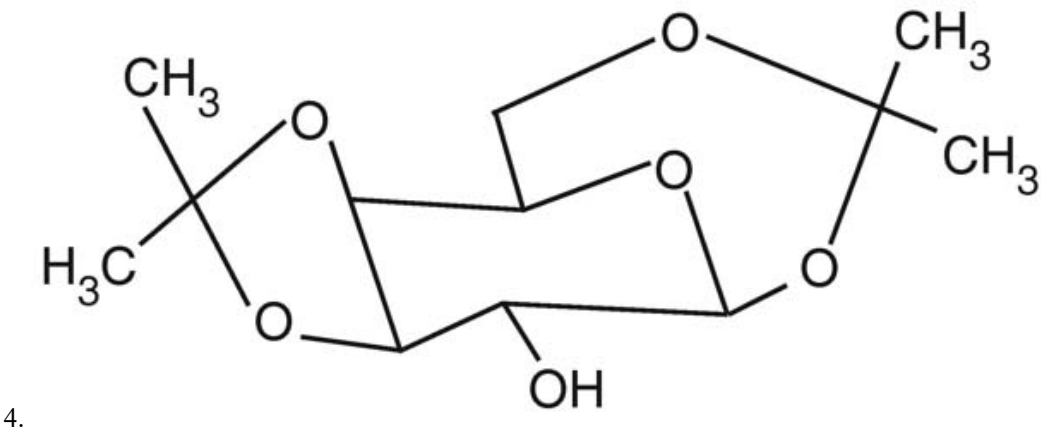
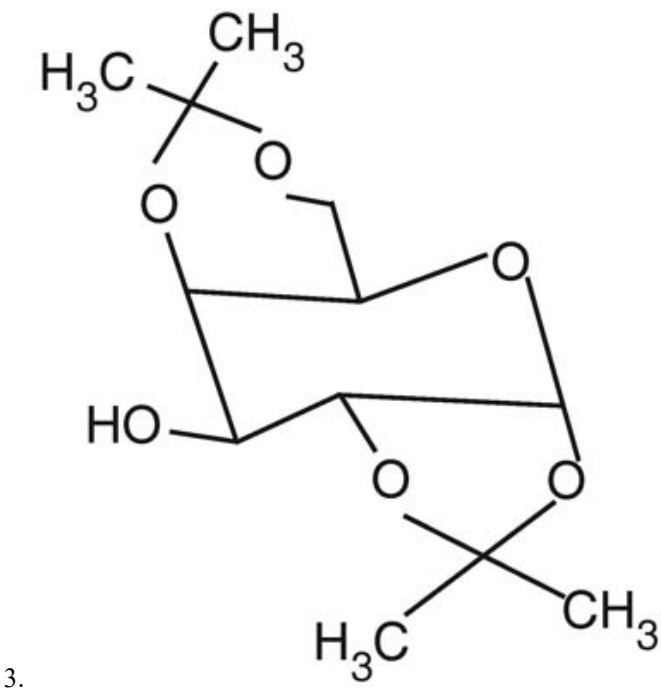
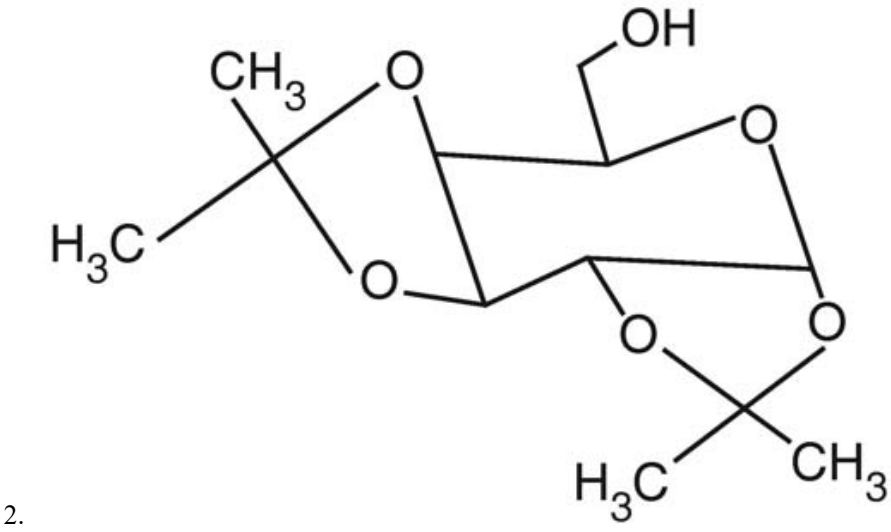
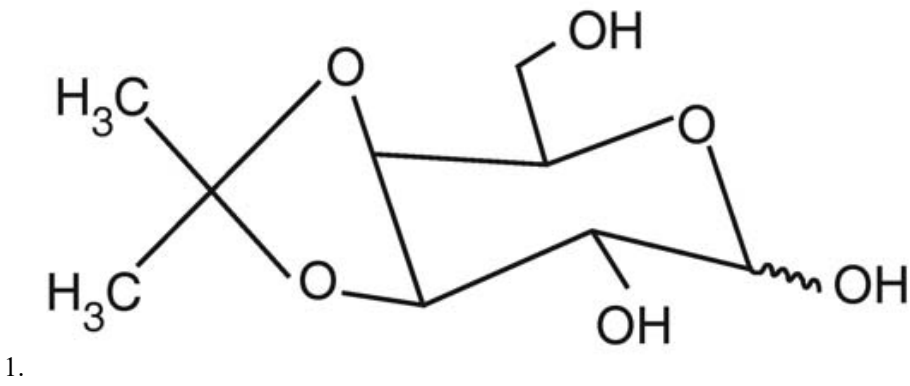
3. H_2SO_4

4. HCl

Reaction of D-galactose with acetone and H^+ gives :-



Options :



Identify the incorrect pair from the following options.

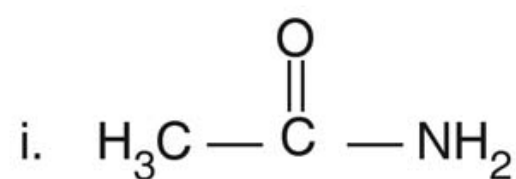
Options :

1. Alkaloid : Codeine
2. Steroid : Cortisone
3. Terpenoid : Menthol
4. Lipid : β -carotene

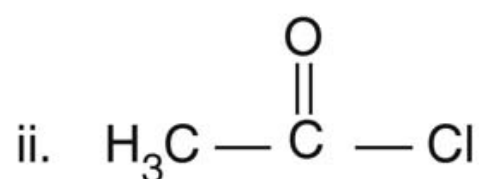
Match the IR carbonyl frequency ($\gamma_c = 0$) of compounds given in column – I with the IR values (cm^{-1}) given in column – II.

Column – I

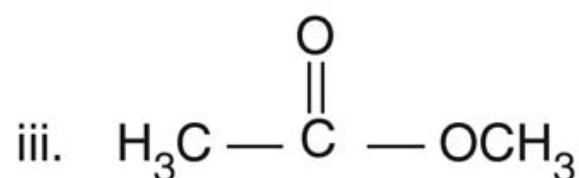
Column – II



a. 1736 cm^{-1}



b. 1694 cm^{-1}



c. 1822 cm^{-1}

Options :

1. i – c, ii – b, iii – a
2. i – b, ii – c, iii – a
3. i – a, ii – c, iii – b
4. i – b, ii – a, iii – c

Which of the following techniques is used for the separation of compounds like petroleum products, saturated and unsaturated hydrocarbons, sulphur compounds and waxes ?

Options :

1. Gas chromatography
2. NMR spectrum
3. Mass spectrum
4. IR spectrum

A compound with molecular formula $C_4H_8O_2$ shows following 1H NMR spectra.

3.6 (s, 3H)

2.3 (q, J = 6Hz, 2H)

1.0 (t, J = 6Hz, 3H)

The structure of the compound is _____.

Options :

1. $H_3C - CH_2 - O - CH_2 - CHO$
2. $H_3C - \overset{\overset{O}{||}}{C} - O - CH_2 - CH_3$
3. $H_3C - CH_2 - \overset{\overset{O}{||}}{C} - O - CH_3$
4. $H_3C - \underset{\underset{CH_3}{|}}{CH} - O - \overset{\overset{O}{||}}{C} - H$

Which of the following is an example of an azo dye ?

Options :

1. Methylene blue
2. Methyl orange
3. Malachite green
4. Rhodamine – B

Nylon-6,6 is synthesized by condensation of _____.

Options :

1. hexamethylenediamine and caprolactam
2. ethylene glycol and terephthalic acid
3. hexamethylenediamine and adipic acid
4. caprolactam alone

Which of the following methods is the primary industrial process for the manufacture of acetaldehyde ?

Options :

1. Stephen reduction of methyl cyanide
2. Rosenmund reduction of acetyl chloride
3. Wacker process for ethylene oxidation
4. Hydration of acetylene

The major drawback of organochlorine pesticide is _____.

Options :

1. high cost
2. low efficiency
3. environmental persistence
4. poor insecticidal action

What is the bye product in the manufacture of phenol from cumene ?

Options :

1. Acetone
2. Ethanol
3. Acetic acid
4. Acetaldehyde

Paracetamol is mainly used as an _____.

Options :

1. Antibiotic
2. Analgesic and antipyretic
3. Antacid
4. Antiseptic

In the catalytic reforming process for BTX production _____.

Options :

1. Naphtha is converted into aromatics.
2. Coal is carbonized to produce BTX.
3. Toluene is oxidized to benzene.
4. Ethylene is polymerized to give xylene.

